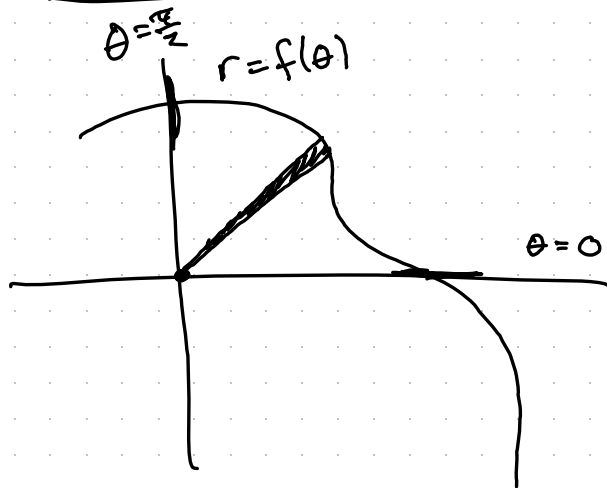
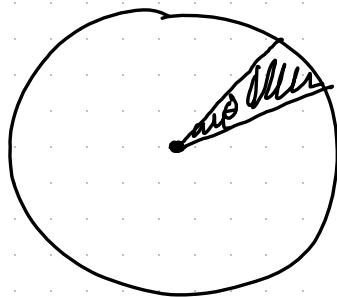


Polar areas — 7.4



Want to add up these little circle segments



A small wedge here contributes

$$\approx \frac{r^2}{2} \theta = \frac{f(\theta_i)^2}{2} \Delta \theta$$

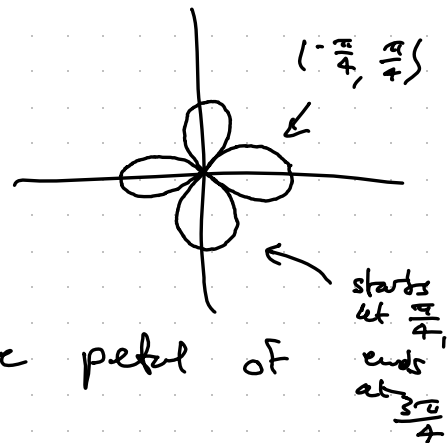
Area $\approx \sum_{i=1}^n \frac{f(\theta_i)^2}{2} \Delta \theta$, which is a Riemann sum!

θ takes up a $\frac{\theta}{2\pi}$ proportion, so we should get this proportion of the total πr^2 area.

$$\text{Area} = \frac{\theta}{2\pi} \cdot \pi r^2 = \frac{r^2}{2} \theta$$

Taking $n \rightarrow \infty$ gives

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{f(\theta_i)^2}{2} \Delta\theta = \int_{\theta_1}^{\theta_2} \frac{f(\theta)^2}{2} d\theta$$

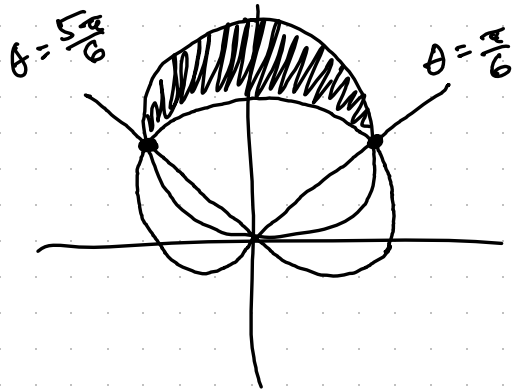


Example: Find the area inside one petal of the curve given by $r = \cos(2\theta)$.

$$\text{Area} = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{2} \cos^2(2\theta) d\theta = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{4} (1 + \cos(4\theta)) d\theta$$

$$\begin{aligned} &= \frac{1}{4} \theta + \frac{1}{16} \sin(4\theta) \Big|_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \\ &= \frac{1}{4} \cdot \frac{\pi}{4} + \frac{1}{16} \sin(\pi) - \frac{1}{4} \left(-\frac{\pi}{4}\right) - \frac{1}{16} \sin(-\pi) \\ &= \frac{\pi}{16} + \frac{\pi}{16} = \frac{\pi}{8} \end{aligned}$$

Example: Find the area inside the circle $r = 3\sin\theta$ but outside the cardioid $r = 1 + \sin\theta$.



When do these intersect?

$$3\sin\theta = 1 + \sin\theta$$

$$2\sin\theta = 1$$

$$\sin\theta = \frac{1}{2}, \quad \theta = \frac{\pi}{6}, \frac{5\pi}{6}.$$

$$\int_{\pi/6}^{5\pi/6} \frac{1}{2} (3\sin\theta)^2 - \frac{1}{2} (1 + \sin\theta)^2 d\theta = \pi.$$

